

ENCT UNIVERSITY, BHOPAL
B.TECH (CSE) VI SEMESTER
Cloud Computing-II

Maximum Marks: 70

Time Allowed: 3 Hours

Note:- Attempt all questions. Internal choice is given.

SECTION – A

I. Short Answer Type Questions (Attempt Any Five) [5×4=20]

- i. Define DevOps and mention its key business drivers.
- ii. What is Continuous Integration (CI) and how is it different from Continuous Delivery (CD)?
- iii. Explain Infrastructure as Code (IaC) with a suitable example.
- iv. What are the benefits of using Jenkins in DevOps pipelines?
- v. Describe the purpose of playbooks and inventory in Ansible.
- vi. What is a Dockerfile? How is it used in building containerized applications?
- vii. Define Kubernetes and list any two of its core components.

SECTION – B

II. Long Answer Type Questions (Attempt Any Four) [4×10=40]

- i. Explain the **DevOps lifecycle** and compare it with **traditional operations**.
- ii. Describe the **DevOps maturity lifecycle** and how it helps in strategic progression.
- iii. Explain how Jenkins integrates with GitHub and Maven. Also, describe how a job is built using Jenkins.
- iv. Discuss the key components and features of **Chef** and **Ansible** in DevOps automation.
- v. Explain the process of creating and managing containers using **Docker** and deploying them via **Kubernetes**.
- vi. Discuss the importance of monitoring in DevOps. Explain how **Nagios** and **Splunk** can be used for system health checks and alerts.